

Mutual Fund Analytics Platform Capstone Project

**Developed By:
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**Technologies Used:
Python | SQL | SQLite | Power BI**

Bluestock Fintech



OVERVIEW

Objective:

To analyze mutual fund industry data and build an interactive analytics dashboard.

Key Focus Areas:

- Fund Performance Analysis
- Investor Analytics
- SIP Trends
- Industry Growth Analysis
- Business Intelligence Dashboard
- Result
- Conclusion



ABSTRACT

The Mutual Fund Analytics Platform is a data analytics and business intelligence solution developed to analyze mutual fund industry trends, fund performance, investor behavior, and SIP growth patterns. The project integrates 10 financial datasets and applies ETL, data cleaning, SQL analytics, and advanced performance metrics such as CAGR, Sharpe Ratio, Sortino Ratio, Alpha, and Beta.

Interactive Power BI dashboards were developed to provide insights into industry growth, fund performance, investor transactions, and market trends. The platform transforms raw financial data into actionable insights that support data-driven investment analysis and decision-making.



PROBLEM STATEMENT

Challenges:

- Large volume of mutual fund data
- Difficult to identify top-performing funds
- Limited visibility into investor behavior
- Lack of consolidated analytics platform

Solution:

Develop a centralized Mutual Fund Analytics Platform.



DATASET OVERVIEW

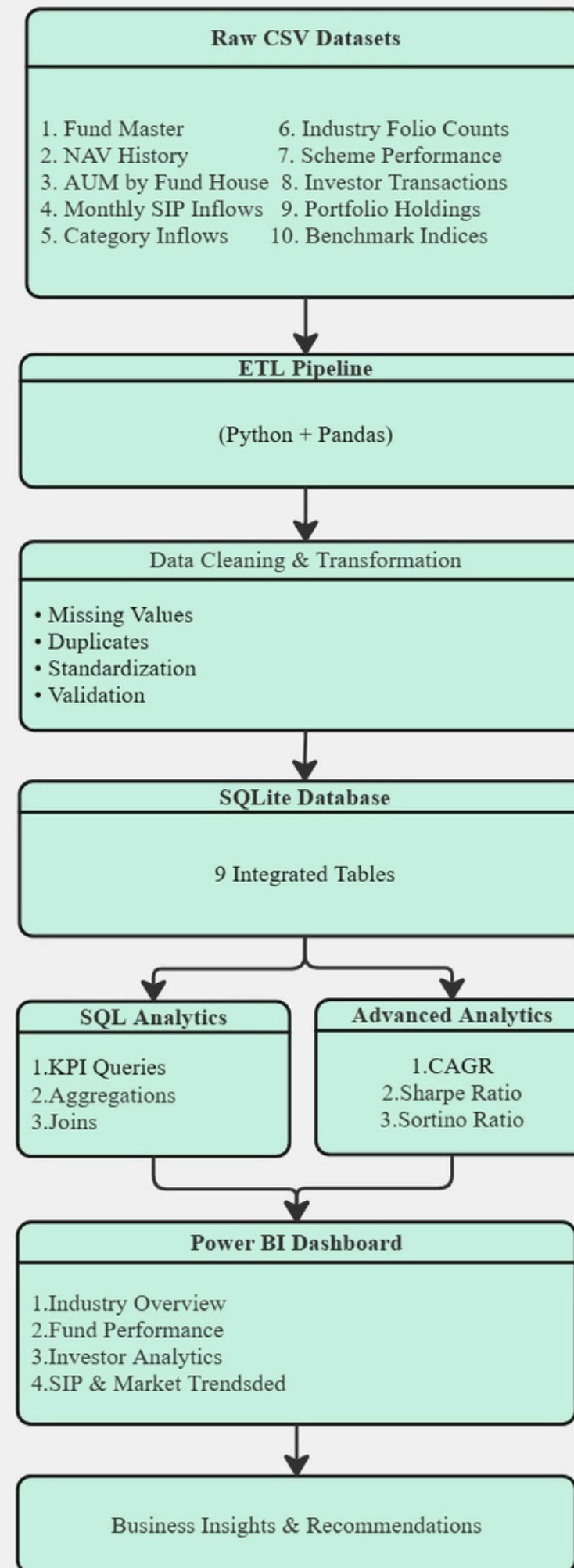
Datasets Used:

- Fund Master
- NAV History
- Performance
- Transactions
- SIP
- AUM
- Category
- Benchmark
- Folio

Total Records Processed:
87,211+



DATA ARCHITECTURE



ETL & DATA CLEANING

- Data Processing Activities:
- Missing Value Handling
- Duplicate Removal
- Standardization
- Validation Checks
- Feature Engineering
- Result:
- Clean and Analysis-Ready Dataset

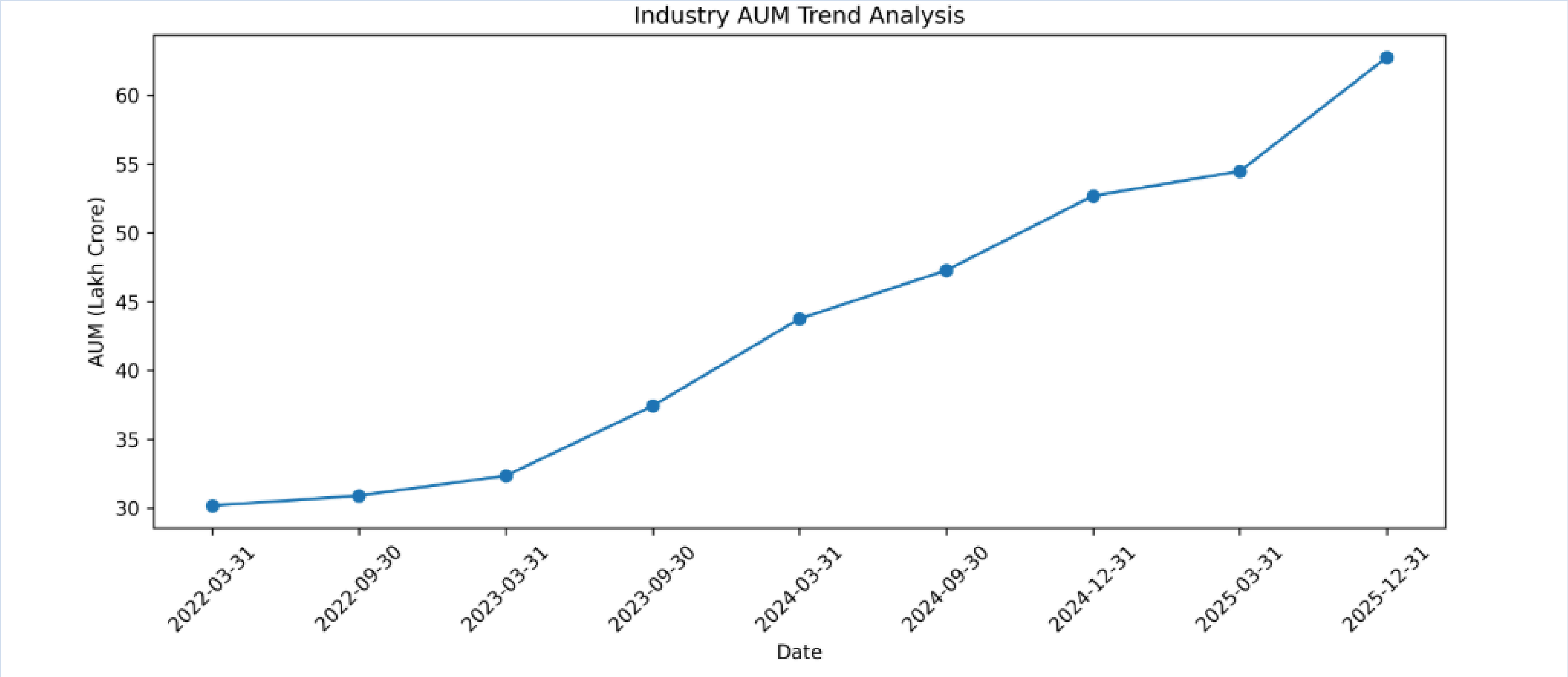


EXPLORATORY DATA ANALYSIS

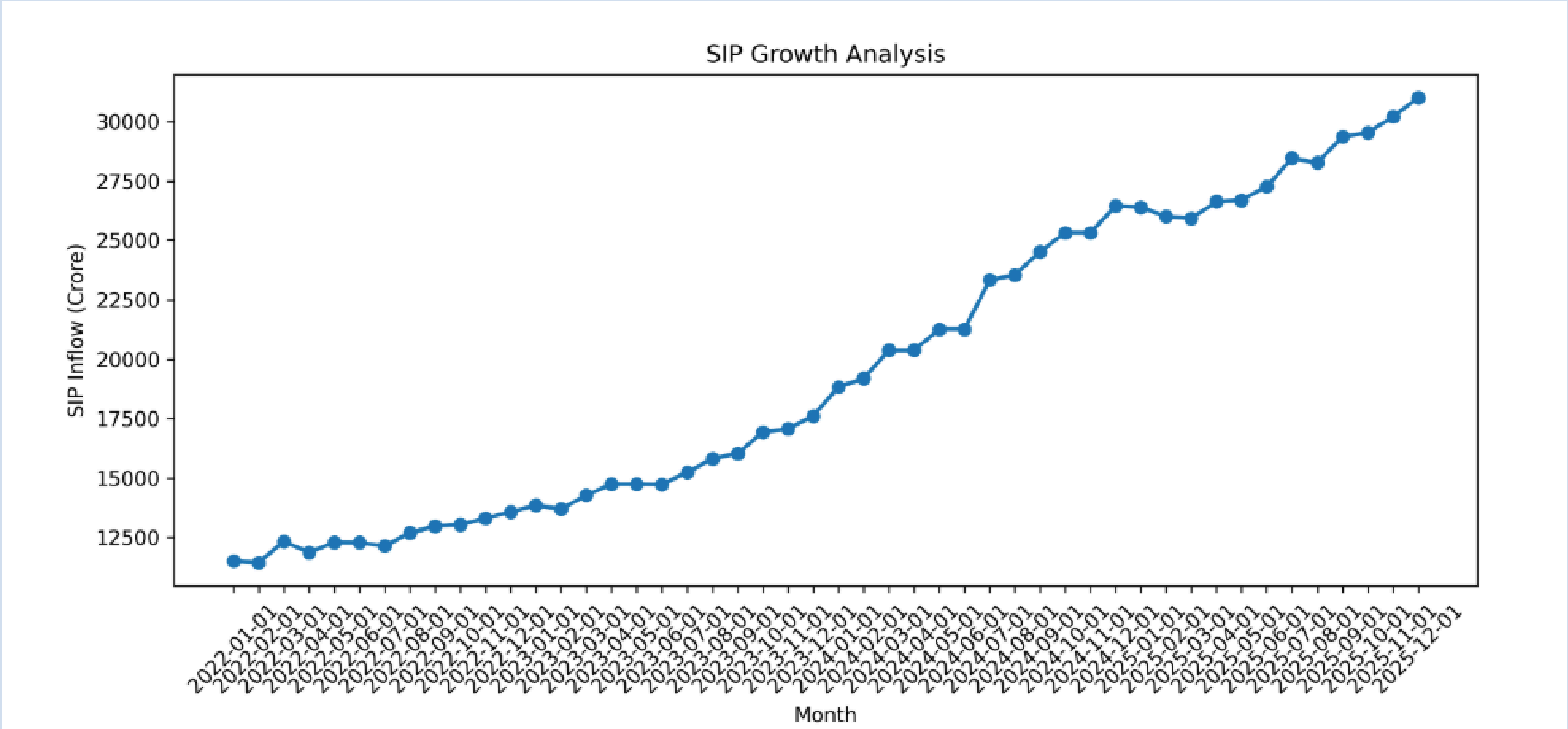
- Industry AUM increased steadily
- SIP participation showed growth
- Category inflows varied significantly
- Investor participation expanded



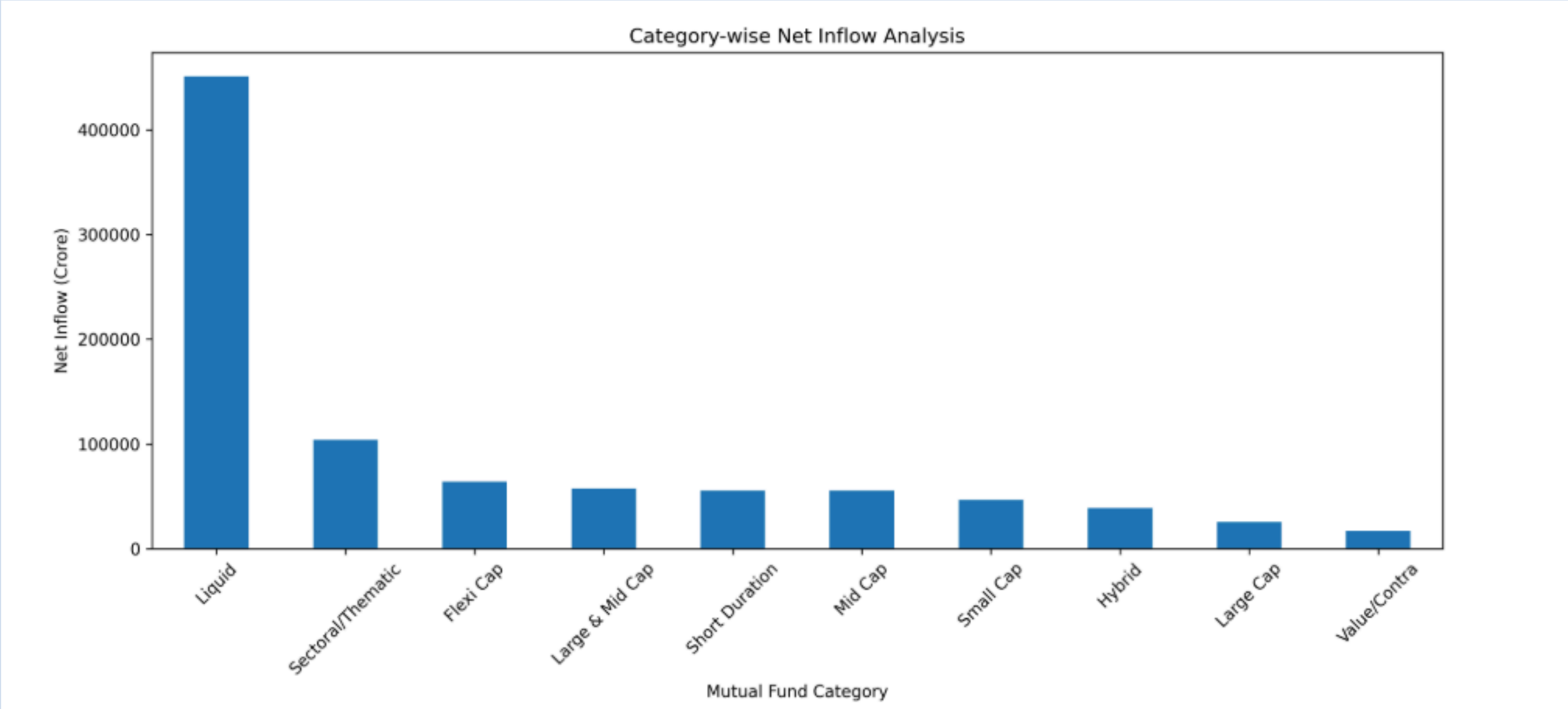
EXPLORATORY DATA ANALYSIS



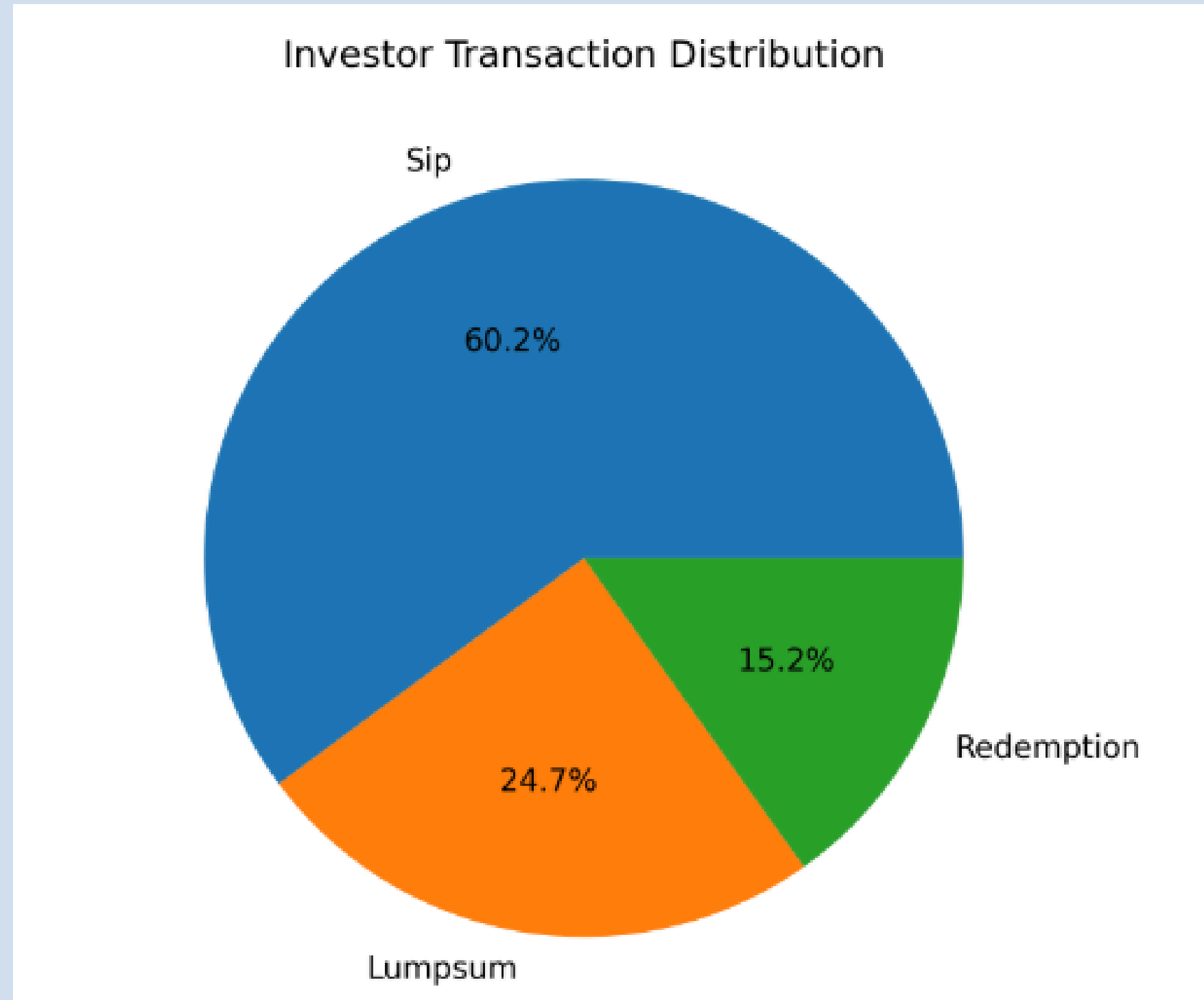
EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS



EXPLORATORY DATA ANALYSIS



PERFORMANCE ANALYTICS

- Industry AUM increased steadily
- SIP participation showed growth
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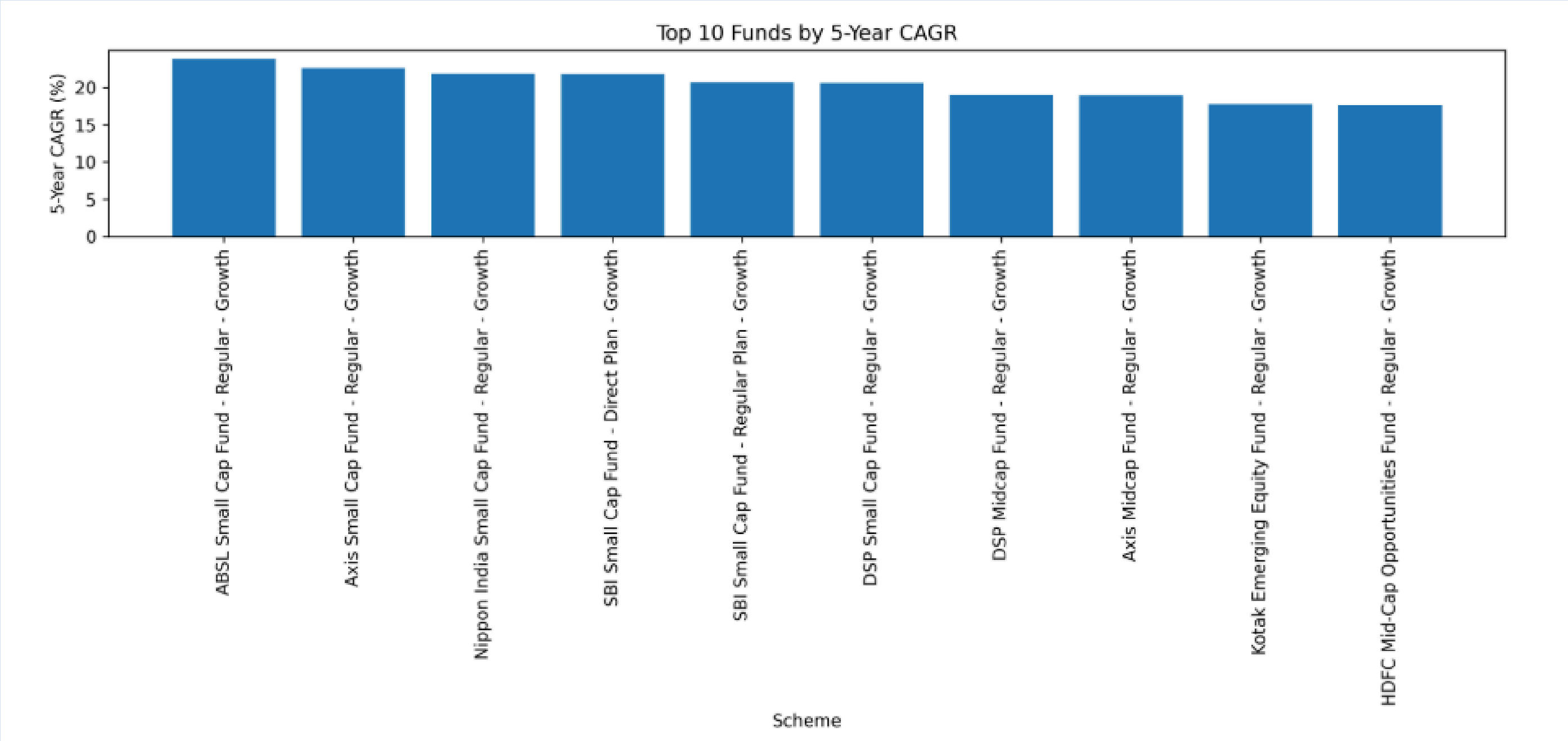
PERFORMANCE ANALYTICS

Metrics Used:

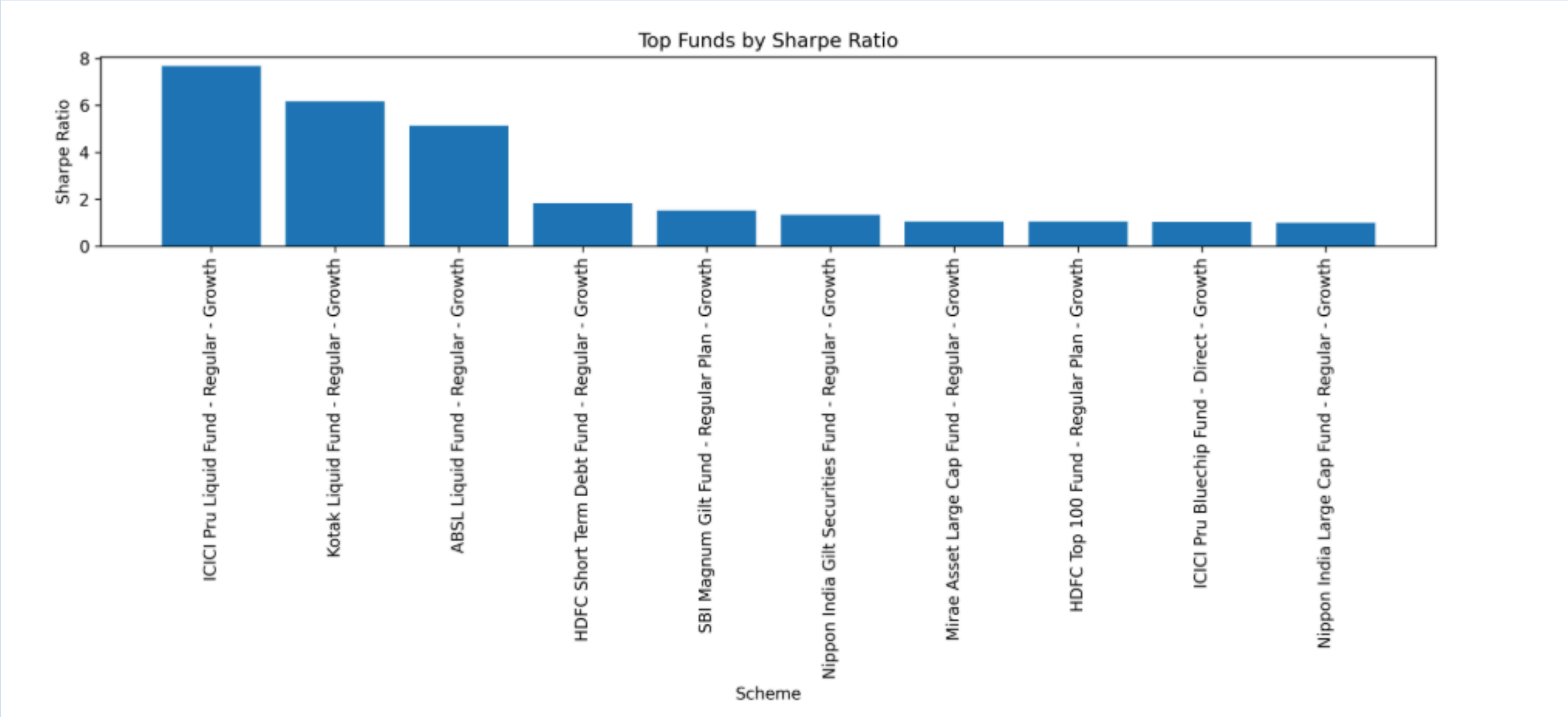
- CAGR
- Sharpe Ratio
- Sortino Ratio
- Alpha
- Beta
- Max Drawdown



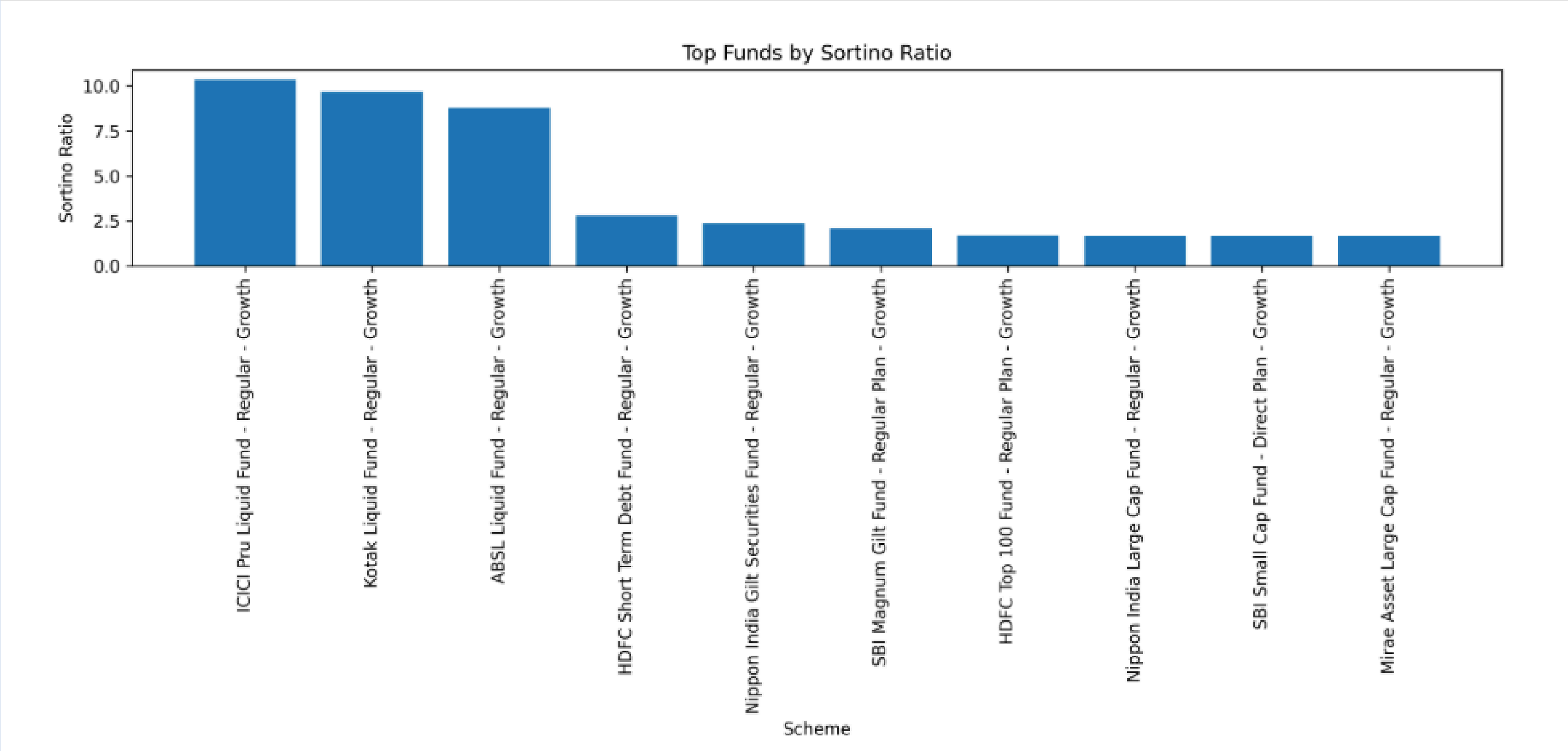
PERFORMANCE ANALYTICS



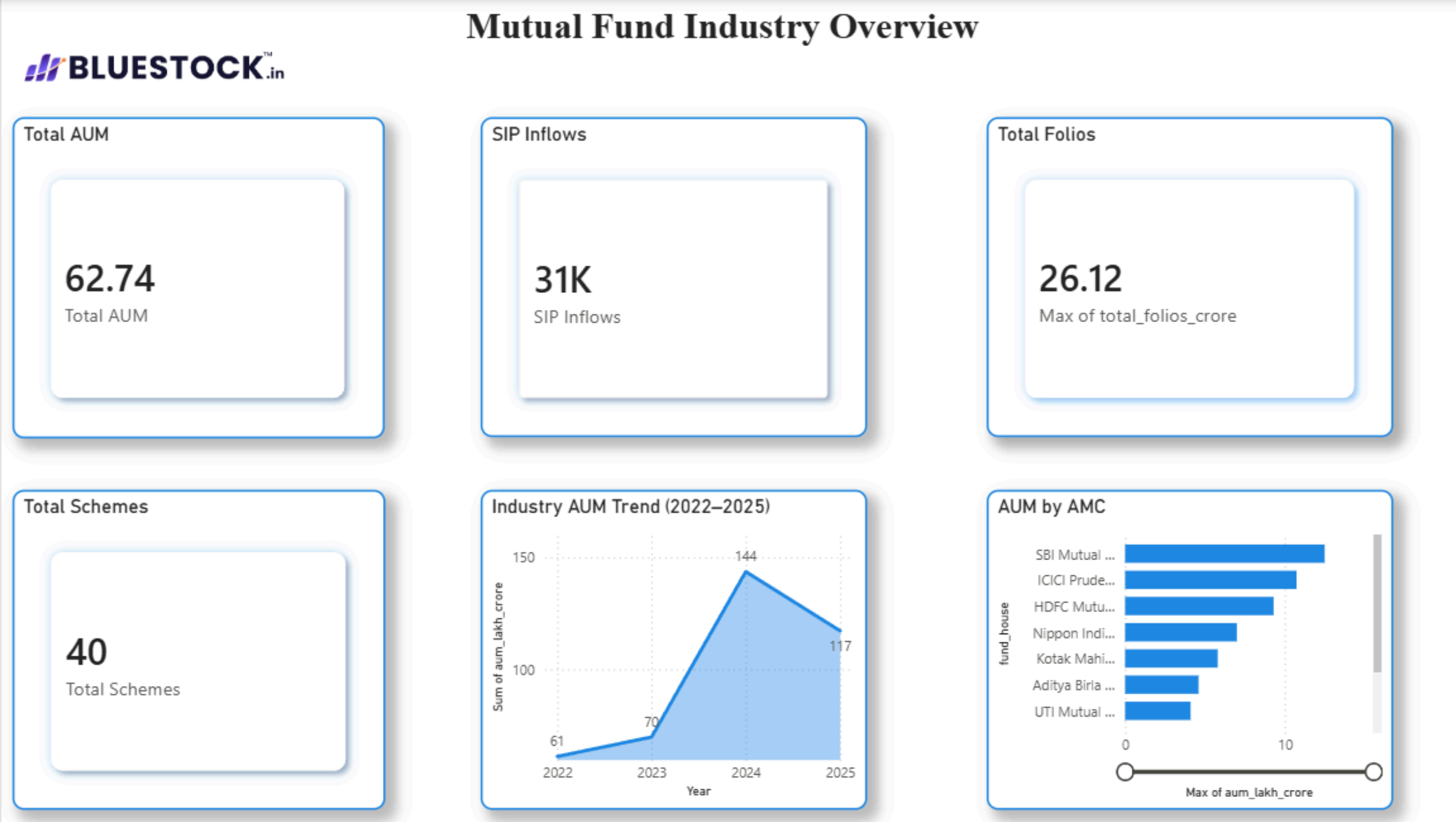
PERFORMANCE ANALYTICS



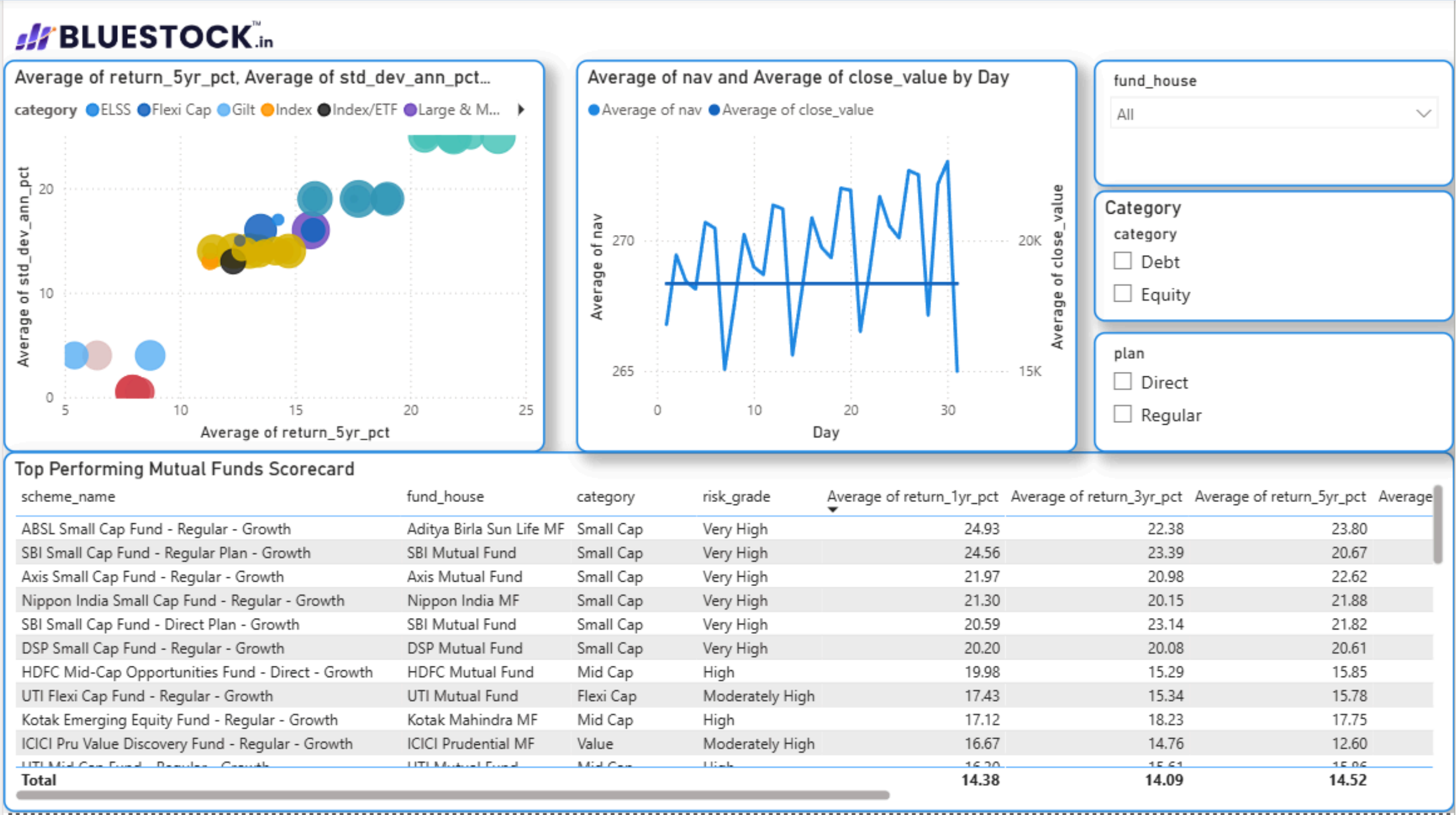
PERFORMANCE ANALYTICS



DASHBOARD INDUSTRY OVERVIEW

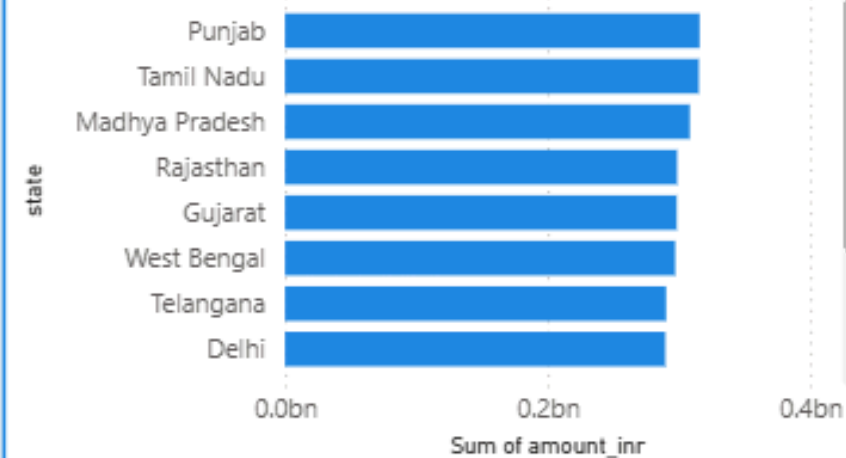


FUND PERFORMANCE

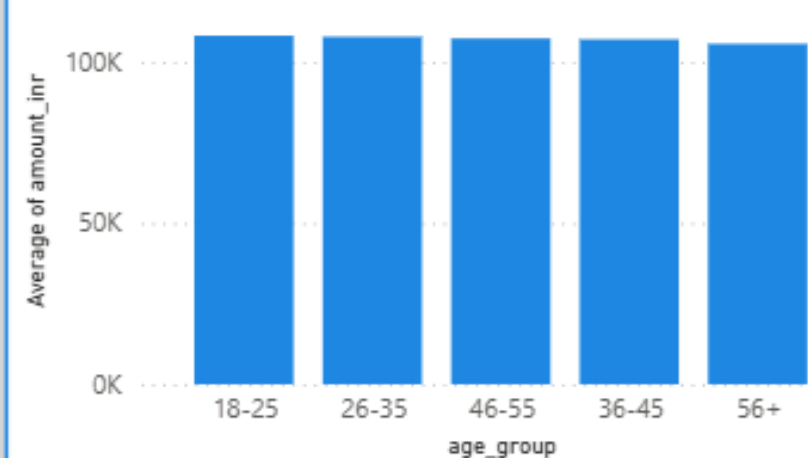


INVESTOR ANALYTICS

Transaction amount by state



Average SIP Amount by Age Group



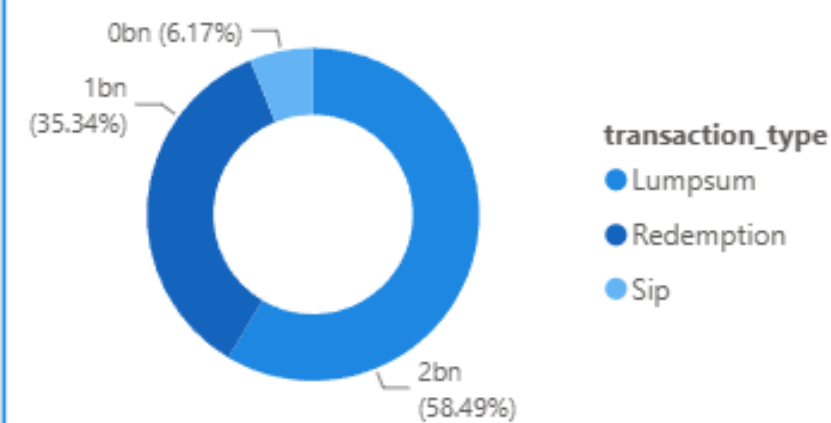
state

- ☐ Delhi
- ☐ Gujarat
- ☐ Haryana
- ☐ Karnataka
- ☐ Madhya Pradesh
- ☐ Maharashtra
- ☐ Punjab
- ☐ Rajasthan

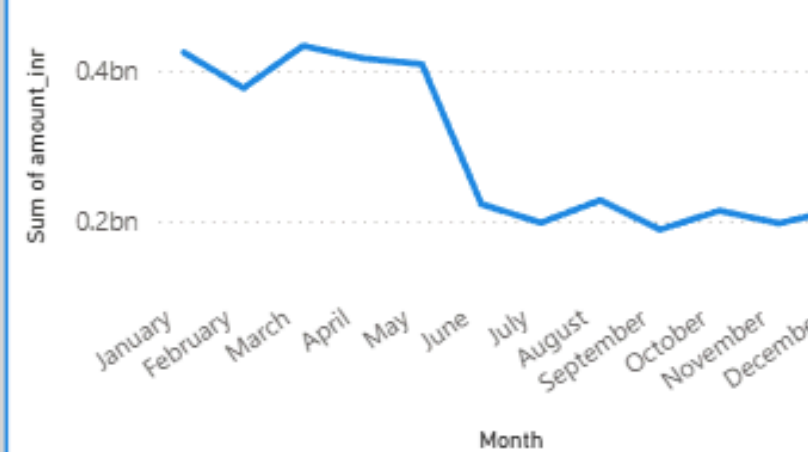
age_group

- ☐ 18-25
- ☐ 26-35
- ☐ 36-45
- ☐ 46-55
- ☐ 56+

Transaction Type Distribution



Monthly Transaction Volume Trend



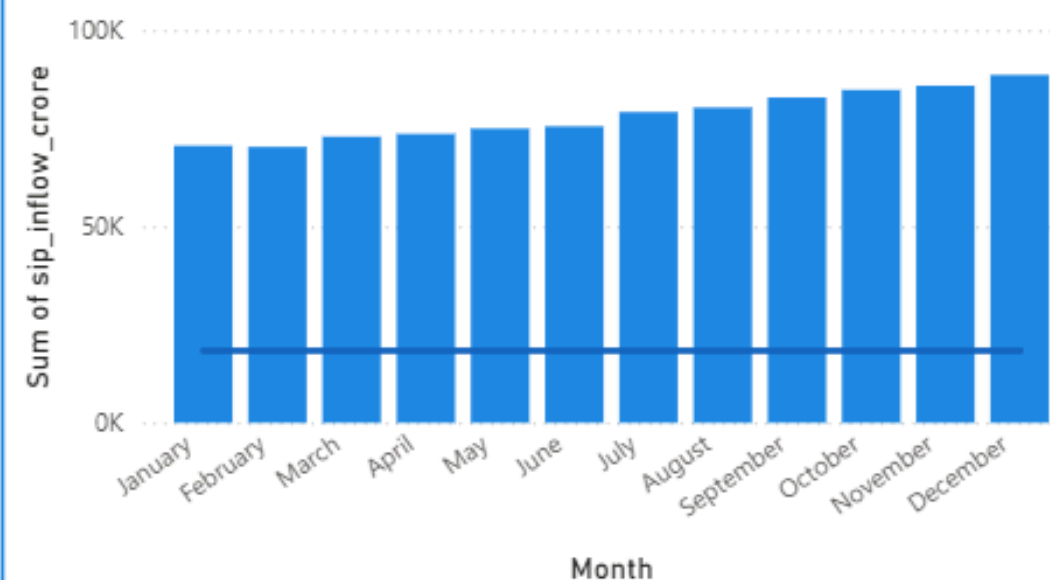
city_tier

- ☐ B30
- ☐ T30

SIP & MARKET TRENDS

SIP Inflows vs Nifty 50 (2022–2025)

● Sum of sip_inflow_crore ● Average of close_value



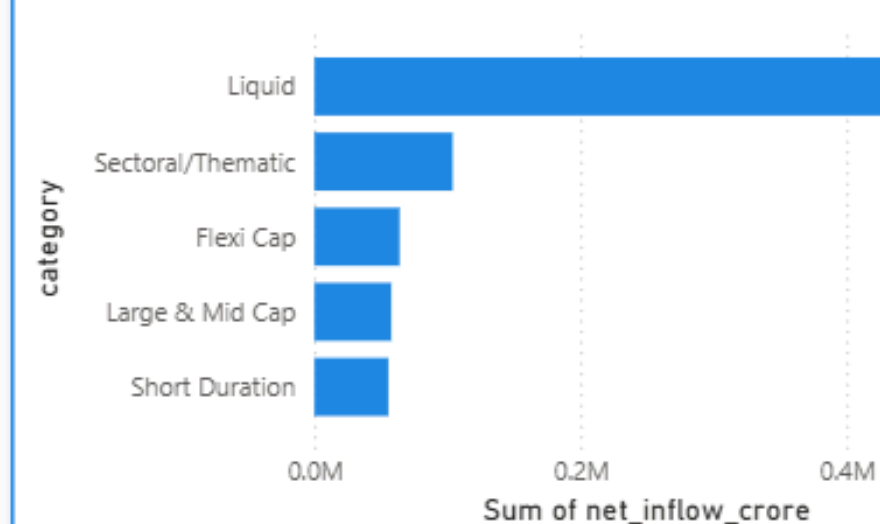
index_name

- ☐ BSE_SMALLCAP
- ☐ CRISIL_GILT
- ☐ CRISIL_LIQUID
- ☐ NIFTY_MIDCAP150
- ☐ NIFTY100
- ☐ NIFTY50
- ☐ NIFTY500

Category Inflow Heatmap.

category	2024	2025	Total
ELSS	4627	1453	6080
Flexi Cap	47551	16438	63989
Gilt	7753	2642	10395
Hybrid	29711	9157	38868
Large & Mid Cap	43169	14583	57752
Large Cap	19449	6184	25633
Liquid	346328	104947	451275
Mid Cap	41116	14196	55312
Sectoral/Thematic	78107	25722	103829
Short Duration	41859	13671	55530
Small Cap	34204	12392	46596
Value/Contra	12744	4236	16980
Total	706618	225621	932239

Top 5 Categories by Net Inflow FY25



KEY FINDINGS

- AUM growth remained strong
- SIP inflows increased consistently
- Investor participation expanded
- Top funds outperformed benchmarks
- Risk-adjusted metrics improved fund evaluation



CONCLUSION

The Mutual Fund Analytics Platform successfully transformed raw financial data into actionable business insights using Python, SQL, SQLite, and Power BI.

The solution enables efficient analysis of mutual fund performance, investor behavior, and industry trends through an interactive dashboard.



Thank You

